Sarah He

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Education

University of California, San Diego September 2023 - June 2026
Bachelor of Science in Data Science *GPA: 3.88*

Experience

Data Analyst Intern June 2025 - August 2025
Capital One *McLean, VA*
- Designed and implemented SQL-based compliance controls, automating previously manual audit procedures and integrating them into a centralized platform
- Diagnosed and resolved issues in legacy compliance logic, removing 85% of false positives and streamlining data validation
- Consolidated fragmented datasets into a centralized data mart to support transparent, consistent reporting and simplified downstream analysis
- Collaborated with cross-functional partners to align control outputs with business logic, ensuring accurate reflection of policy requirements and compliance goals

Power Transformation Lab Research Assistant November 2025 - August 2026
UC San Diego *La Jolla, CA*
- Engineered robust Python ETL pipelines for international trade data, monitoring live data ingestion and proactively diagnosing production anomalies through logging frameworks
- Leveraged AI coding assistants to optimize API data extraction architecture, implementing dynamic batching algorithms and exponential backoff retries
- Collaborated with senior researchers on database automation initiatives supporting low-carbon energy and climate research

Data Science Intern August 2024 - February 2025
A Round Entertainment *Remote*
- Developed and optimized predictive models (CatBoost, LightGBM and Ensemble methods) to estimate concert revenue, capturing over 90% of the patterns in the data and uncovering key factors influencing revenue generation
- Developed interactive Tableau dashboards that allowed non-technical stakeholders to explore data by artist and time period, effectively visualizing key metrics and trends
- Scraped and parsed HTML data using the BeautifulSoup library to build robust datasets with 705k+ rows from online sources, enabling in-depth analysis of concert industry trends
- Automated attendance tracking by setting up new employee attendance system in Excel, reducing manual effort and increasing efficiency in performance analysis

Co-Founder June 2020 – September 2023
Starry Sticker Shop *Temple City, CA*
- Managed product design, customer service, and financial planning, growing the Etsy business to 270+ global orders
- Analyzed and visualized sales patterns and customer sentiment for actionable insights using Python and Tableau

Projects

Intuit Virtual Expert Platform Workforce Optimization | *Python (Pandas), Time Series, Erlang-A, UI Integration*
- Developed and fine-tuned a dual-horizon demand forecasting pipeline using GradientBoostingRegressors to predict call volumes at 30-minute intervals, achieving a 3.11% WMAPE
- Integrated an abandonment-aware Erlang-A queueing emulator to translate volume predictions into strict operational headcount requirements and SLA targets
- Implemented frontend functionality for an interactive Performance Simulator dashboard, enabling workforce managers to dynamically test SLA targets against predictive staffing models

Major Power Outage Analysis | *Python (Pandas, Scikit-learn, Matplotlib), HTML, CSS*
- Conducted in-depth analysis of power outage data to identify regional and infrastructural factors influencing outage severity
- Applied feature engineering and hyperparameter tuning to develop a Logistic Regression model that forecasts severe power outages with 85% accuracy
- Conducted permutation testing to assess missingness dependency, model fairness, and analyze key trends in the dataset

EcoMenu | *Python (Pandas, Matplotlib, Flask), Node.js, Tableau, HTML, CSS, JavaScript*
- Sourced, compiled, and cleaned datasets on the carbon footprint of over 350 food ingredients and 2100 recipes
- Developed web application enabling users to select ingredients and receive low-carbon recipe recommendations
- Presented visualizations to showcase insights from the recipe and ingredient data
- Boost Hacks II: Most Creative Hack, CirFin CREATE: Civic Achievement Award

Skill Set

Technical: Python, SQL, Tableau, Power BI, Excel, Snowflake, Git, Java, R, HTML/CSS, JavaScript, Jupyter Notebook, Figma, Flask
Production: Adobe Lightroom, Adobe Photoshop, CapCut, PowerPoint